

Kapitza Pendulum

Numerical Assignment Classical Physics

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1 Introduction

A Kapitza pendulum is a setup where the pivot point has its own oscillation, either vertically or horizontally. Such setups make for an interesting study as their behavior differs a lot from the classical pendulum. This Assignment's focus is derivation of the analytical solution with use of Lagrangian mechanics and numerical simulation of the motion using the Verlet Algorithm.

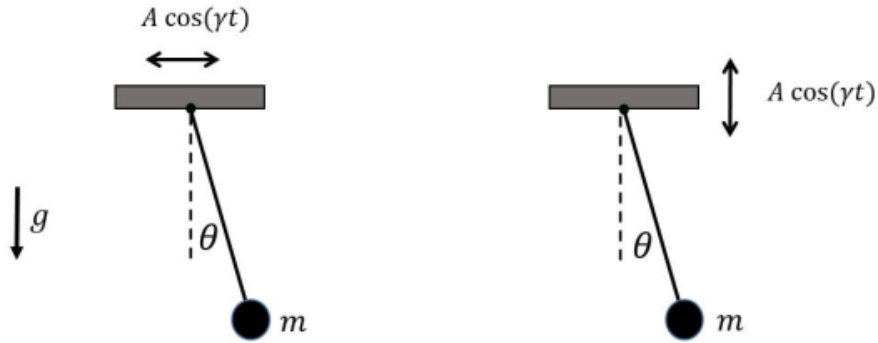


Figure 1: Schematic drawing of the Kapitza pendulum with oscillating pivot.

2 Analytical Solution

2.1 Horizontal Kapitza Pendulum

We begin by modifying our coordinates x and y in the following manner:

$$x = l \sin(\theta) + A \cos(\gamma t) \quad y = -l \cos(\theta)$$

Next, we find the Lagrangian for our system using kinetic and potential energies.

$$T = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2) \quad V = -mgl \cos(\theta)$$

$$L = T - V = \frac{1}{2}ml^2\dot{\theta}^2 + \frac{1}{2}mA^2\gamma^2 \sin^2(\gamma t) - mlA\gamma\dot{\theta} \sin(\gamma t) \cos(\theta) + mgl \cos(\theta)$$

Using Lagrangian equation for the equation above we end up with an expression of a harmonic oscillator with some damping factor.

$$\ddot{\theta} = \frac{A\gamma^2}{l} \cos(\gamma t) \cos(\theta) - \frac{g}{l} \sin(\theta)$$

If we choose to study small initial angles θ , the damping term reduces to $\frac{g}{l}\theta$ while at the same time $\cos(\theta)$ simply becomes 1, making this differential equation trivial for solving analytically.

2.2 Vertical Kapitza Pendulum

The process for the vertical pendulum will be largely similar, with some terms changing to their trigonometric counterparts.

$$x = l \sin(\theta) \quad y = -l \cos(\theta) + A \cos(\gamma t)$$

$$L = T - V = \frac{1}{2}ml^2\dot{\theta}^2 + \frac{1}{2}mA^2\gamma^2 \sin^2(\gamma t) + mlA\gamma\dot{\theta} \sin(\gamma t) \sin(\theta) + mgl \cos(\theta) + mgA \cos(\gamma t)$$

If we now apply the Lagrangian equation we still end up with a damped harmonic oscillator:

$$\ddot{\theta} = \frac{A\gamma^2}{l} \cos(\gamma t) \sin(\theta) + \frac{g}{l} \sin(\theta)$$

However, if we now apply the small angle approximation $\sin(\theta) = \theta$ the differential equation we are left with is much more troublesome.

2.3 General Solution to the Horizontal Kapitza Pendulum

We take the non-homogeneous differential equation we derived earlier and rearrange the terms so that the left side has all the functions of θ , while the right side depends on t .

$$\ddot{\theta} + \frac{g}{l}\theta = \frac{A\gamma^2}{l} \cos(\gamma t)$$

We start by ignoring the right side and treat the ODE as if it was homogeneous. This will simply be the general solution for all harmonic oscillation:

$$\theta_h = A \sin(\sqrt{g/l}t) + B \cos(\sqrt{g/l}t)$$

where A and B are some constants while $\sqrt{g/l}$ is the new angular frequency ω_0 . Since the right side is a trigonometric function we make an educated guess that the particular solution should be:

$$\theta_p = C \cos(\gamma t) + D \sin(\gamma t)$$

Now we substitute the particular solution into the left side of the differential equation and end up with restrictions for the equation to hold:

$$-C\gamma^2 + \frac{g}{l}C = \frac{A\gamma^2}{l} \Rightarrow C = \frac{A\gamma^2}{g - l\gamma^2} = \frac{A\gamma^2/l}{\omega_0^2 - \gamma^2}$$

$$-D\gamma^2 + \frac{g}{l}D = 0 \Rightarrow D = 0$$

The General Solution to the horizontally oscillating Kapitza pendulum becomes this:

$$\theta = A \sin(\sqrt{g/l}t) + B \cos(\sqrt{g/l}t) + \frac{\tilde{C}\gamma^2}{\omega_0^2 - \gamma^2} \cos(\gamma t)$$

There is also the possibility of $\gamma = \sqrt{g/l}$ that would also satisfy the condition for coefficient D . This implies $C = 0$ and arbitrary D . We study this case in more detail in the numerical part.

3 Numerical Analysis

We will now solve the two differential equations acquired earlier using numerical integration. Firstly, since both the ODEs are of second order we save effort by converting each into two 1st order ODEs.

$$\dot{\theta} = v_\theta$$

$$\dot{v}_\theta = \frac{A\gamma^2}{l} \cos(\gamma t) \cos(\theta) - \frac{g}{l} \sin(\theta) \equiv F[\theta, t]$$

Here we defined the right side as its own function with two inputs - time and angle.

The method we will be using is called the Verlet Algorithm which bases itself on the more general midpoint method. The underlying idea is picking some starting values and for a chosen small time delimiter dt we update the old value at dt in two steps with first the value that corresponds to $t + dt/2$ and then the value that corresponds to $t + dt$. The procedure for updating angle and angular velocity values will therefore look like this:

$$v_\theta(t + \frac{\Delta t}{2}) = v_\theta(t) + F[\theta(t), t] \frac{\Delta t}{2}$$

$$\theta(t + \Delta t) = \theta(t) + v_\theta(t + \frac{\Delta t}{2}) \Delta t$$

$$v_\theta(t + \Delta t) = v_\theta(t + \frac{\Delta t}{2}) + F[\theta(t + \Delta t), t] \frac{\Delta t}{2}$$

3.1 Global Error Assessment

The explicit midpoint method is a 2nd order method, meaning the global error is proportional to the delimiter squared.

$$\|\epsilon_N\| = \|\theta(t_N) - \theta_N\| \leq C\Delta t^2$$

A method of second order is by no means the best option, but with an appropriate choice of Δt we can minimise the global error to suit our purposes i.e. produce detailed enough data sets and graphs that explain the pendulums motion. We can illustrate this further by studying the total energy over some interval of time. For simplicity we can ignore the damping factors and study conservation of energy for a simple pendulum which is given by following expression:

$$E = \frac{1}{2}ml^2\dot{\theta}^2 + mgl(1 - \cos(\theta)) - \text{const.}$$

Since there is no damping we should in theory have energy conservation, but due to accumulation of local error for each step, we will end up with some variation. If we run the simulation for different starting angles we notice that the variation in total energy becomes more significant the larger the angle is. The best way to counteract this issue would be to increase number of steps or decrease dt . The relative error in the data set can be expressed like this:

$$\delta E = \frac{\max(E_T) - \min(E_T)}{E(\dot{\theta}_0, \theta_0)}$$

where we use the initial values to calculate the true error-free energy. For $N = 500$ the relative error is around 0.4%, which is acceptable, but this value drops below 0.1% if we double the number of steps to $N = 1000$.

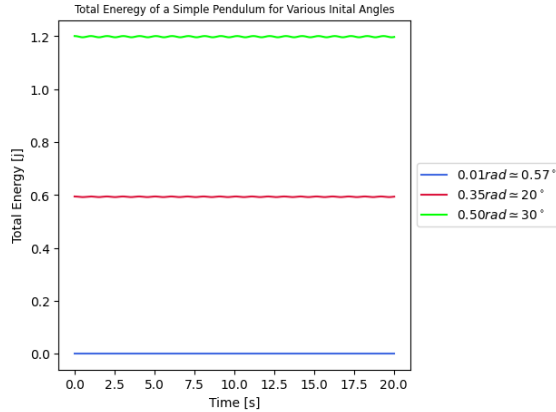


Figure 2: Energy conservation for a simple pendulum. The affect of numerical error becomes more apparent for larger initial angles.

4 Simulation Results

4.1 Slight Damping

We start to analyse the Kapitza pendulum by studying small pivot oscillations $\gamma = 0.2\omega_0$, $A = 0.05L$. We notice that both the angle and angular velocity graphs are already different from that of the simple pendulum. The Horizontal graph has a shape similar to a bending accordion, while the vertical is the same oscillator but with amplitude that keeps increasing with time. Since the damping effect is set to be small, the pendulum will maintain some of its previous behaviour and the equilibrium position is still around 0. We also notice that halving Δt does not impact the result.

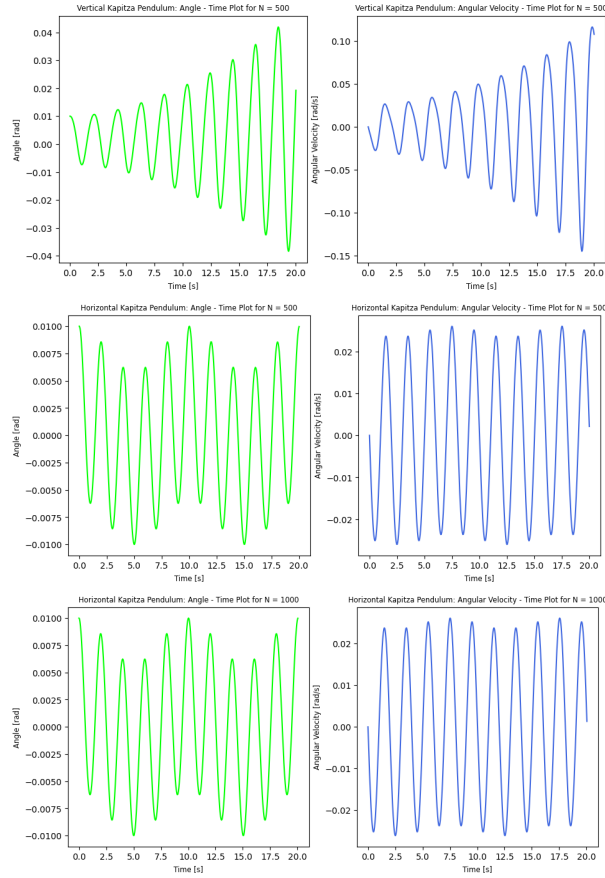


Figure 3: Angle and angular velocity graphs for a pendulum with small pivot oscillations (horizontal/vertical). Bottom graph shows that halving the step size does not make a difference.

4.2 Resonance Dynamics

We will now make the pivot oscillation more impactful by increasing its oscillation frequency $\gamma \in [0.1\omega_0, 3\omega_0]$. We choose to plot maximal amplitude of the angle and the total energy for both the horizontal and vertical cases. If we let N be sufficiently large (around 1000) we get a clear spike somewhere around $\omega_0 \approx \sqrt{g/L}$ for the horizontal pendulum. Remember how the analytical solution we derived earlier had the term $\frac{\tilde{C}\gamma^2}{\omega_0^2 - \gamma^2} \cos(\gamma t)$. If we let γ approach (but remain smaller than) ω_0 this term will blow up in value which is exactly what we are observing graphically. For the vertical Kapitza pendulum the resonant pivot frequency lies somewhere below $2\omega_0$.

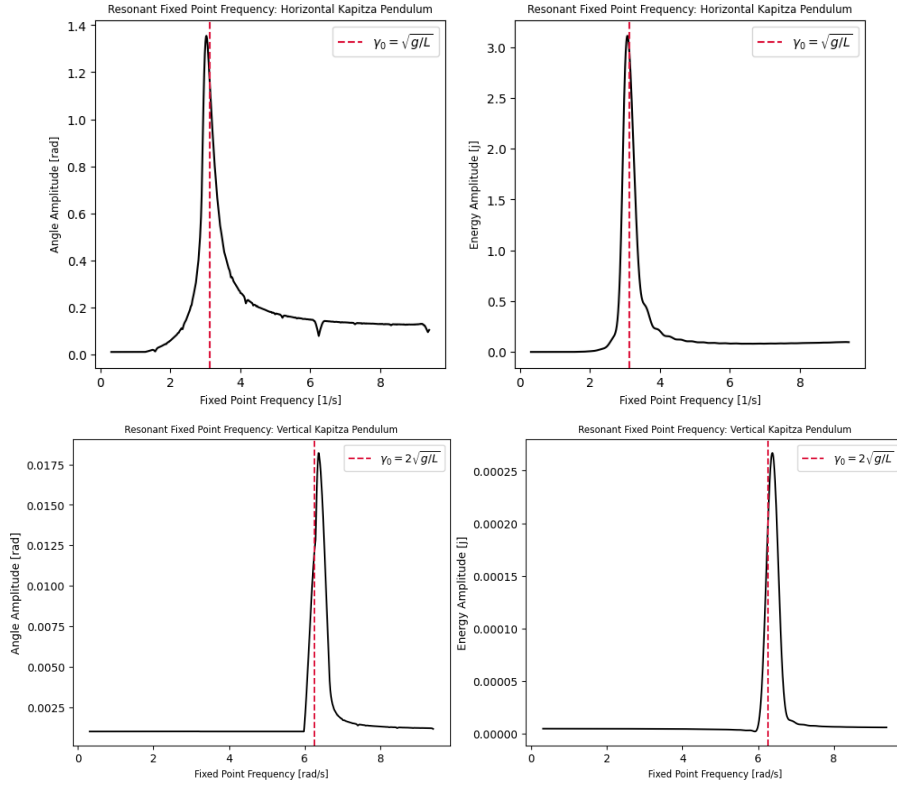


Figure 4: Maximum angle amplitude and total energy as function of the pivot frequency γ , reveal the resonance spikes around ω_0 and $2\omega_0$ for horizontal and vertical Kapitza pendulums respectively.

4.3 Equilibrium Positions

What makes the Kapitza pendulum so special are the new equilibrium positions that become available due to the complex motion. Below we simulate this motion for different starting angles $\theta = [0, \pi/4, \pi/2, \pi]$ and for pivot frequencies much larger than previously, $\gamma = [10\omega_0, 50\omega_0, 100\omega_0, 200\omega_0]$.

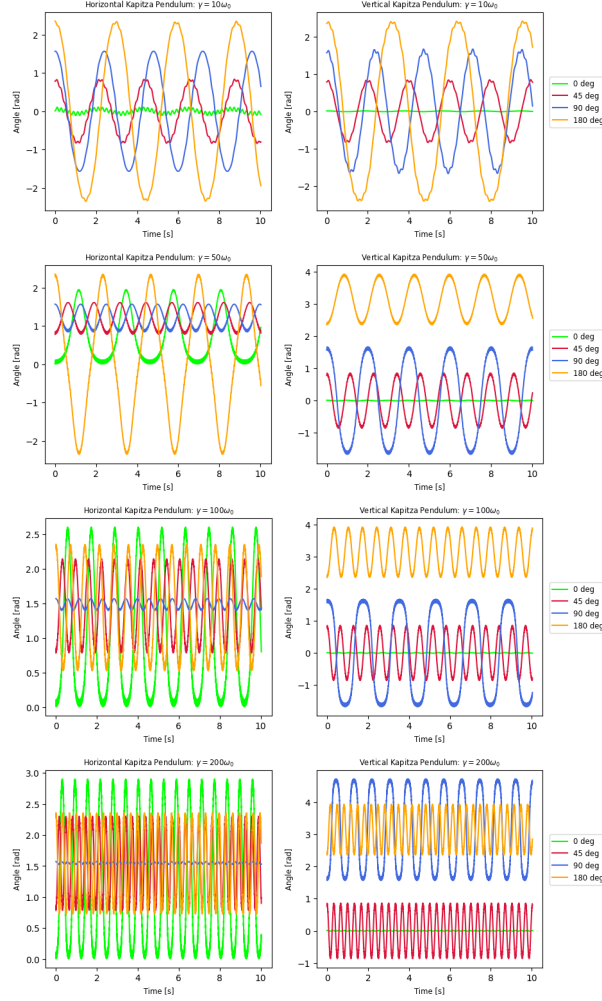


Figure 5: Equilibrium positions for horizontal and vertical pivot oscillations. The horizontal gains a new equilibrium around $\pi/2$, while the vertical Kapitza pendulum gains π . Both are impossible for the regular pendulum.

In case of horizontal oscillation a new equilibrium is established around $\pi/2$ at frequencies $\gamma \geq 50\omega_0$, replacing the old state around 0. The horizontal oscillation creates a force that counteracts the pull of gravity. This new force prohibits the pendulum from falling back to the old equilibrium and essentially allowing the mass to stay perpendicular to the gravitational field. This becomes more apparent for very high frequencies, where the pendulum remains above zero during the entire time interval. At lower frequencies the counter force is weak and we observe a little bit of both behaviours (for large angles the old equilibrium remains).

The vertical Kapitza pendulum gains a new equilibrium position around π . This means the pendulum mass stays essentially anti-parallel with the gravitational field. In order for the pendulum to enter the higher equilibrium it needs a large angle. The higher the oscillation frequency of the pivot, the lower the initial angle can be in order for the pendulum to start oscillating around the top position.

5 Conclusion

This project's focus was on deriving and solving the Lagrangian equations for the horizontal and vertical Kapitza pendulum and backing this up with numerical simulation. This was done using the explicit midpoint method. We showed both resonance behaviour and equilibrium positions for various pivot frequencies. The resulting resonance frequencies were in agreement with the analytical predictions. Finally, the numerical model illustrated how an oscillating pivot allows the Kapitza pendulum to gain motion properties that would be impossible for a simple pendulum.